JianghuSect 2 (JS2-v2.6.0) — Deep-Dive Feature & Logic Audit

System #01 (Part 2 of 2): Cultivation Paths & Subpaths — Progression Math, 4-Wave Parity, Dead Peak Geometry, Era-5 Dormancy, God-Script Elimination & Future-Proof Schema

- **Target Codebase**: Js2 new (SIM.PATCH_VERSION = '2.6.0', SIM.SAVE_SCHEMA_VERSION = '2.6.0-living-chronicle-and-reporting', 43 Major Paths / 286 Subpaths)
- **Target Readers**: Autonomous Coding Models (**GPT** & **Gemini** Systems Architects / Developers)
- **Prerequisite Reading**: 01A_JS2_Deep_Audit_Paths_and_Subpaths_Part1_Lifecycle_Selection_Retraining (Sections 0–5: Core Emergent Philosophy, Locked human_core.js 155:2f9c9dd7, Genesis/Path Library, Trait/Constitution ×62 Inversion, Subpath Alias Collisions, and Retraining Lifecycle).
- **Audit Scope (Part 2 of 2 — Sections 6 to 10)**:
  - **Section 6 ([PATH-MATH-06])**: Asymmetric maxStage (4 / 5 / 8 / 9) XP/Lifespan/Power Distortion, Unimplemented GOD Faith-Collapse Counterweight, effectiveTechStage Track-Grinding Tier-Skip Exploit, and subpathTraits curOrd Ternary Bug.
  - **Section 7 ([PATH-WAVE-07])**: Four-Wave Historical Inequality Across Daily AI, **80 Unreachable Sacred Peak Stations**, Shrine Bottlenecking of 31 Paths, Mantra/hpMult Bias, and Reverse-Nerf in COMBAT.strike.
  - **Section 8 ([PATH-ERA5-08])**: 100% Dormancy of Era-5 Overhaul Subsystems (MANTRA_SYSTEM, DEFENSIVE_ARTS, SUPPORT_ARTS, FORBIDDEN_ARTS) in Autonomous Play & Elimination of Player Puppeteering Controls.
  - **Section 9 ([PATH-GDAO-09])**: Disembodied Monthly God-Scripts (monthlyAutonomousTick), -76% Pantheon Pact Sign Inversion, Split-Brain Runtime Stances (MECH_PUPPET / KARMA_SAMSARA), and SHADOW_SAMURAI Zero-Qi Edge Cases.
  - **Section 10 ([PATH-FUT-10])**: Declarative PATH_BEHAVIOR_DESCRIPTOR Architecture (Eliminating Scattered if (path === ...) Chains for Future Scalability).

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6. [PATH-MATH-06] Asymmetric maxStage (4 / 5 / 8 / 9) Progression Distortion, effectiveTechStage Exploit & subpathTraits Masking Bug

6.1 Exact Code Anchors

- data/data_cult.js: XIAN, BODY, GOD, DEMON, HEART_SOUL stage tables
- js/cultivation.js: CULT.THRESH (lines 726–747), CULT.effectiveTechStage (lines 750–757), CULT.normalizedStage (lines 759–765), CULT.hpMult (lines 767–772)
- js/stats.js: compute & getBaseFast subpathTraits evaluation (lines 318–334, 461–479)

6.2 Root-Cause Code & Mathematical Defects

Defect 6A — 38.5× Cumulative Cultivation XP Disparity & Missing GOD Faith-Collapse Counterweight

Compare the cumulative cultivation progress (CULT.THRESH) required to reach the Apex Stage across the 4 stage structures in cultivation.js:

Path Group	maxStage	Breakthrough Array (CULT.THRESH)	Cumulative XP to Apex	Apex Base combatPower	Apex lifespanMax
GOD	4	[300, 1200, 5200]	**6,700 XP**	**32,500**	**2,000 yrs**
BODY	5	[230, 800, 3200, 12000]	**16,230 XP**	**31,000**	**1,500 yrs**
38 Paths (Waves 2–4)	8	[~250, ~680, ~1740, ~4400, ~10800, ~26400, ~63500]	**~107,770 XP**	**~28,000–31,000**	**~1,600–1,900 yrs**

Path Group	`maxStage`	Breakthrough Array (`CULT.THRESH`)	Cumulative XP to Apex	Apex Base `combatPower`	Apex `lifespanMax`
`XIAN` / `DEMON` / `HEART_SOUL`	`9`	`[120, 320, 850, 2200, 5600, 14000, 35000, 85000, (150k)]`	**~143,090–258,000 XP**	**30,000–31,000**	**1,800–1,950 yrs**

- **Why This Breaks Simulation Balance & Lore Context**:
  1. In `data/data_cult.js`, `GOD` was designed with only 4 stages because its lore explicitly specifies brutal external world gates and existential risk:
     - Stage 1 `resourceReq`: *"ศาลเทพประจำเมืองและผู้ศรัทธา 1 หมื่นคน"*
     - Stage 3 `resourceReq`: *"เครือข่ายศาลเทพ 3 เมืองและศรัทธา >= 78"*
     - `coreMechanicTh`: *"หากเมืองในปกครองล่มสลายหรือผู้คนหมดศรัทธาจะตกขั้นทันที!"*
  2. **Actual Code Reality**: A `GOD` disciple sits inside the sect `shrine` (`mult = 1.25`), gains full `CULT.rate` **boosted** further by `1 + ritual * 0.0035` (`proficiency.js:116`), needs **zero** external cities or believers, has **zero** stage-loss risk from faith collapse, and reaches **32,500 base combat power & 2,000-year lifespan in just 6,700 XP**—the exact same XP where a `XIAN` disciple has barely entered Stage 5 (`combatPower = 2,250`, `lifespan = 360`)!
  3. **Long-Horizon Ripple Effect**: Over a 15-year autonomous simulation, any sect or rival faction with `GOD` (or `BODY` at `16,230 XP`) produces Immortal-tier apex juggernauts 10×–25× faster than 8-stage and 9-stage paths, completely warping World Martial Conferences, Siege wars, and Sect Tier rankings.

Defect 6B — `CULT.effectiveTechStage` Track-Grinding Tier-Skip Exploit (`cultivation.js:750–757`)

JavaScript

```
CULT.effectiveTechStage = n => {
  if (!n || !n.cult) return 0;
  const c = n.cult, st = c.stage || 1;
  const trOrd = (c.track && c.track.ordinal) || 0;
  let mapped = st;
  if (c.path === "BODY") mapped = [1, 2, 4, 6, 7][Math.min(4, st - 1)] || st;
  else if (c.path === "GOD") mapped = [1, 3, 5, 7][Math.min(3, st - 1)] || st;
```

```
  else if ((CULT.PATHS[c.path] && CULT.PATHS[c.path].maxStage === 8)) mapped = [1,
2, 3, 4, 5, 6, 7, 7][Math.min(7, st - 1)] || st;
  return Math.max(st, mapped, trOrd);
};
```

- **Critical Logic Flaw**: Including raw `trOrd` (`1..9`) inside `Math.max(st, mapped, trOrd)` decouples equipment/manual tier eligibility from Major Realm cultivation (`c.stage`)!
  - Specialization tracks (`c.track`) have **no Heavenly Tribulation**, **no lifespan bottleneck**, and early ordinals (`200`, `520`, `1350`, `3400`) advance steadily via skill drills and passive gain.
  - Consequently, a **Stage 1 Qi-Refining outer disciple** (`c.stage = 1`, `qiMax ≈ 80`) whose `c.track.ordinal` reaches `6` or `7` acquires `effectiveTechStage = 7`—allowing a Stage-1 mortal-realm novice to **equip Tier-7 Primordial Divine Relics (`minStage: 7`), buy Tier-6/7 gear from `CPE.shop`, and learn Tier-7 Heaven-Shattering Legendary Arts (`reqTier: 7`)** without ever forming a Foundation or Golden Core!
  - Also note `martial.js:244` (`canActivateMantra`): `if (reqT > effSt && (c.path !== 'GOD' || reqT > 2)) return false;`—an arbitrary legacy exception letting Stage-1 `GOD` activate Tier-2 mantras for free.

Defect 6C — `subpathTraits` `curOrd` Ternary Masking & Unreachable `reqRealm: 9` (`stats.js:320, 463`)

Every one of the 286 subpaths defines 3 milestone `subpathTraits` at `reqRealm` **3, 6, and 9**. Yet `stats.js` computes:

```JavaScript
const curOrd = (c.track && c.track.ordinal)
  ? c.track.ordinal
  : ((c.path === 'BODY' || c.path === 'GOD') && CULT.normalizedStage ?
Math.floor(CULT.normalizedStage(n)) : c.stage);
```

2. **Masking Bug**: Whenever `c.track` is initialized (`{ ordinal: 1 }`), `(c.track && c.track.ordinal)` is truthy (`1`). Thus, a Stage 8/9 Elder (recruited from wandering figures, generated at sect creation, or promoted) whose `c.track.ordinal` is `1` evaluates `curOrd = 1`, unlocking **zero** `subpathTraits` despite being at the apex of cultivation!
3. **Unreachable `reqRealm: 9` on 38 Paths**: When `!c.track` on any of the **38 paths with `maxStage === 8`**, the fallback only calls `CULT.normalizedStage` for `BODY` and

`GOD`—returning raw `c.stage` (max `8`) for all 38 `maxStage = 8` paths, so `curOrd >= 9` is **mathematically impossible**!

6.3 Architectural Remediation Blueprint for AI Developers

1. **Re-Calibrate `GOD` & `BODY` Stage Thresholds + Genuine `GOD` Faith Mechanics**:
   - Scale `CULT.THRESH.GOD` and `CULT.THRESH.BODY` (or path-specific `CULT.rate` pacing factor) so reaching Stage 4 `GOD` / Stage 5 `BODY` requires comparable real simulation months (`~65%–85%` of an 8-stage path's total time-to-apex), **and** tie `GOD` breakthrough gates & cultivation rate to real sect/vassal/regional faith (`W.sect.fame`, `shrine` worship activity, allied/vassal settlements, and public relief actions), applying a temporary *"Shaken Faith (ศรัทธาสั่นคลอน)"* cultivation/power penalty if sect fame/morale collapses after catastrophic defeats.
2. **Bound `trOrd` Contribution in `CULT.effectiveTechStage`**:
   - Replace `Math.max(st, mapped, trOrd)` with `Math.max(st, mapped) + (trOrd >= mapped + 2 ? 1 : 0)` (allowing exceptional subpath mastery to bridge at most **+1 tier** above the cultivator's Major Realm equivalent, never letting a Stage-1 novice jump 6 tiers to Tier-7 Divine Relics!).
3. **Fix `curOrd` in `stats.js` (`compute` & `getBaseFast`) and `CULT.GRAND_DAO_MECHANICS.getActiveTraits`**:
   - Compute `const normStage = CULT.normalizedStage ? Math.floor(CULT.normalizedStage(n)) : c.stage;` (which maps `maxStage = 4, 5, 8, 9` onto the `[1..9]` scale for **all** paths!) and use `const curOrd = Math.max((c.track && c.track.ordinal) || 0, normStage);` (or synchronize `c.track.ordinal` floor when generating high-stage NPCs).

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7. `[PATH-WAVE-07]` Four-Wave Historical Stratification Across Daily AI, 80 Dead Peak Stations, Stat Curves & Combat Multipliers

7.1 Architectural Context: Why 31 Paths Behave Like Second-Class Citizens

Across patches `v1.0` -> `v2.6.0`, the 43 Major Paths were added in **4 waves**:

- **Wave 1 (5 Paths)**: `XIAN`, `BODY`, `GOD`, `DEMON`, `HEART_SOUL`
- **Wave 2 (7 Paths)**: `MECH_PUPPET`, `KARMA_SAMSARA`, `BEAST_SOVEREIGN`, `SAGE_SUPPORT`, `CRAFT_CREATION`, `ARRAY_TALISMAN`, `STRATEGY_GOVERNANCE`

- **Wave 3 (15 Paths, `v2.0.0`)**: `ASTRO_DIVINATION` .. `PRIMORDIAL_CHAOS`
- **Wave 4 (16 Paths, `v2.2.0`)**: `SHADOW_SAMURAI` .. `ARENA_GLORY`

Multiple core loops in `ai.js`, `bottleneck.js`, `cpeco.js`, `stats.js`, `cultivation.js`, and `combat.js` still hardcode **only Wave 1 & Wave 2 (10–12 paths)**, dumping Waves 3 & 4 (31 paths = **72% of all paths!**) into broken `else` clauses.

7.2 Root-Cause Code & Logic Defects

Defect 7A — CRITICAL MAP/AI BUG: All 80 Sacred Peak Meditation & Practice Stations Are 100% Unreachable Dead Geometry (`js/ai.js:121–145`)

Inspect `AI.freeStation(W, n, kind, bkeys)` in `js/ai.js`:

```javascript
if (typeof bkeys === 'string') {
  const b = W.map.byId[bkeys] || (W.map.buildings.find(x => x.key === bkeys));
  if (b) { /* searches ONLY central building b.stations */ }
  if (bkeys === 'canteen' && kind === 'eat' && n && n.peakId && g.SECT &&
g.SECT.PEAKS) {
    /* checks peak pantry ONLY for eating! */
  }
  return best;
}
```

Meanwhile, in `map.js` (`PEAK_LAYOUTS`), every single awakened Sacred Peak (`PEAK_AZURE_SWORD`, `PEAK_VAJRA_BODY`, `PEAK_MYRIAD_CAULDRON`, etc.) constructs **`6 practice` + `4 meditate` + `2 pantry` physical stations** on that peak (`10` cultivation/training stations × `8` Peaks = **80 stations**)!

- **Why They Never Work**:
  - `A.cultivate.where` calls `AI.freeStation(W, n, 'meditate', 'meditation_hall')`.
  - `A.practice.where` calls `AI.freeStation(W, n, 'dummy', 'training_ground')`.
  - Neither passes the Peak building ID, and `AI.freeStation` **only** redirects `'canteen'` (`'eat'`) to `n.peakId`—**never `'meditate'` or `'practice'`/`'dummy'`/`'spar'`!**
- **Multi-Order Chain Reaction**:
  1. Disciples assigned to a Peak (`n.peakId`) never cultivate or practice on their own Sacred Peak!
  2. In a mid/late-game sect of 50–80 NPCs, central `meditation_hall` and `training_ground` saturate at 100% occupancy (`occ !== null`).

3. Overflowing Elders and Peak Masters fall back to sitting on their dormitory beds (`mult: 0.85` instead of `1.35+` + Peak Sanctuary array bonus!) while **80 purpose-built Peak Sanctuary stations sit completely deserted**!

Defect 7B — Absurd Building Routing for 31 Wave-3/Wave-4 Paths (`ai.js:349–354, 471`, `bottleneck.js:116`, `cpeco.js:337`)

2. **`A.cultivate.where` (`ai.js:349–354`)**: Lumps 23 unrelated paths (`SHADOW_SAMURAI`, `ARENA_GLORY`, `DEVOURING_FLESH`, `NETHER_GRAVE`, `ANCESTRAL_RITUAL`, `LORE_ARCHIVIST`, `SPIRIT_HUSBANDRY`, `FORTUNE_GAMBLER`, etc.) into `[meditation_hall, spirit_pond, forge(array)]`:
    - `SHADOW_SAMURAI` and `ARENA_GLORY` warriors walk to the **Blacksmith Forge array table** to cultivate instead of `training_ground`!
    - `ANCESTRAL_RITUAL` ignores the **Ancestral Hall (`ancestral`)**!
    - `LORE_ARCHIVIST` ignores the **Scripture Pavilion (`scripture`)**!
    - `SPIRIT_HUSBANDRY` ignores the **Herb Garden (`herb_garden`)**!
    - `NETHER_GRAVE` and `VOID_HERETIC_PACT` ignore the **Ancient Ruin (`ruin`)**!
2. **`A.breakthrough.where` (`ai.js:471`), `consolidate.where` (`bottleneck.js:116`), & `paidTraining.where` (`cpeco.js:337`)**:
    - All three functions check **only the 12 Wave-1/Wave-2 paths** and end with `: 'worship'` at `'shrine'`!
    - Every single cultivator across **all 31 Wave-3/Wave-4 paths** (`SHADOW_SAMURAI`, `DEVOURING_FLESH`, `VOID_HERETIC_PACT`, `INK_LITERATI`, `CHRONO_EPOCH`, `FORTUNE_GAMBLER`, etc.) tries to break through, consolidate bottlenecks, and do paid training at the tiny **`shrine` (`worship` station)**—causing severe station starvation and blocking actual `GOD` / `ANCESTRAL_RITUAL` / `DIVINE_*` priests from using the shrine!

Defect 7C — `A.cultivate.tick` Side-Growth & Heart-Demon Starvation on 33 Paths (`ai.js:383–449`)

During `cultivate.tick`, only 10 Wave-1/Wave-2 paths receive path-authentic skill practice, body/mind attribute growth, pet bonding, or `heartDemon` regulation. All 31 Wave-3/Wave-4 paths fall into `else { trainMind('focus'); gainWisdom(0.004); }`:

- **`BUDDHA_DHARMA` (วิถีพุทธะโพธิสัตว์)** gets **zero** `heartDemon` cleansing during meditation, whereas `CRAFT_CREATION` (smiths) and `STRATEGY_GOVERNANCE` (administrators) cleanse `-0.035` to `-0.040` `heartDemon` every 30 ticks!
- **`PRIMORDIAL_SHIFTER`, `DEVOURING_FLESH`, `ARENA_GLORY`, `SHADOW_SAMURAI`** gain zero `strength`/`toughness`/`agility`/`BODY_REFINING`/weapon practice while cultivating, growing into frail scholars instead of physical powerhouses!
- **`SPIRIT_HUSBANDRY` & `VENOM_GU`** receive zero `n.pets` bond/morale ticks during cultivation (unlike `BEAST_SOVEREIGN`).

Defect 7D — Mantra Stat Mismatch (`stats.js:289, 429`), `CULT.hpMult` Bias (`cultivation.js:768`), & Reverse-Nerf in `COMBAT.strike` (`combat.js:1197`)

1. **Mantra Stats (`stats.js:289–301, 429–441`)**: Only 11 paths have custom Mantra stat profiles; all 32 remaining paths get `+4% mdef, +3% hp, +2% matk` per tier (`+0% atk, +0% def, +0% move, +0% crit`)—granting useless `matk` to `SHADOW_SAMURAI` (Zero-Qi physical katana!), `ARENA_GLORY`, `PRIMORDIAL_SHIFTER`, and `NINPO_SHADOW`!
2. **`CULT.hpMult` (`cultivation.js:768`)**: Grants `+0.05 * (ns - 1)` extra HP scaling to Wave-2 backline scholars (`SAGE_SUPPORT`, `ARRAY_TALISMAN`, `STRATEGY_GOVERNANCE`), while giving **`0`** to Wave-3/4 frontline juggernauts (`DEVOURING_FLESH`, `PRIMORDIAL_SHIFTER`, `GRAVITY_DAO`, `ARENA_GLORY`, `BUDDHA_DHARMA`, `DIVINE_POSSESSION`).
3. **Reverse-Nerf in `COMBAT.strike` (`combat.js:1197`)**:

```JavaScript
else if (['ASTRO_DIVINATION', 'DRAGON_VEIN_GEOMANCY', 'LORE_ARCHIVIST',
'MERCANTILE_DAO'].includes(a.cult.path)) {
  frac *= 1 + C.pathFormationMastery(a) * 0.08;
}
// ... whereas the final `else` on line 1207 gives:
else { frac *= 1.09; }
```

   `C.pathFormationMastery(a)` measures `ARRAY` and `TALISMAN` skills. Because `LORE_ARCHIVIST` trains `SCRIPTURE` and `MERCANTILE_DAO` trains `ABACUS`/`trading` (`ARRAY = 0, TALISMAN = 0`), their multiplier is **`1.00x` (`+0%`, max `1.08x` even at 100 skill!)**—strictly **worse** than the unconditional `1.09x` (`+9%`) given to paths in the generic `else`!

7.3 Architectural Remediation Blueprint for AI Developers

1. **Activate All 80 Sacred Peak Stations in `AI.freeStation` (`js/ai.js`)**:
   - Whenever `n && n.peakId && g.SECT && g.SECT.PEAKS` has an awakened peak building `pb` on `W.map`:
     - If `bkeys === 'meditation_hall' && kind === 'meditate'`, check `pb`'s `'meditate'` stations **first** (with Peak Sanctuary `mult` bonus)!
     - If `bkeys === 'training_ground' && (kind === 'dummy' || kind === 'spar' || kind === 'lift' || kind === 'practice')`, check `pb`'s `'practice'` stations **first**!

2. **Map All 43 Paths to Authentic Stations, Cultivation Side-Growth, `hpMult`, Mantra Profiles & Combat Identities** via the unified `PATH_BEHAVIOR_DESCRIPTOR` table in **Section 10**.

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8. `[PATH-ERA5-08]` 100% Dormancy of Era-5 Overhaul Subsystems in Autonomous Play & Elimination of Player Puppeteering

8.1 Exact Code Anchors

- `js/cultivation.js`:
  - `CULT.MANTRA_SYSTEM` (`ensureState` line 1165, `setMantraStance` line 1185, `stepMeditationResonance` line 1210)
  - `CULT.DEFENSIVE_ARTS` (`ensureState` line 1375, `setDefensivePosture` line 1395, `evaluateHitMitigation` line 1425)
  - `CULT.SUPPORT_ARTS` (`ensureState` line 1586, `setSupportPosture` line 1605, `stepFacilityAndDailySynergy` line 1630)
  - `CULT.FORBIDDEN_ARTS` (`ensureState`, `setForbiddenSealMode`)
- `js/ai.js`: `A.cultivate.tick` (lines 378–379)
- `js/combat.js`: lines 765, 811, 871, 916
- `js/ui.js`: Direct posture/stance/seal buttons on NPC sheet

8.2 Root-Cause Code & Design-Philosophy Defects

Defect 8A — All Four Era-5 Subsystems Are 100% Dead Code in Natural Autonomous Runs!

Look at how `ai.js` and `combat.js` invoke the four Era-5 subsystems:

```javascript
// js/ai.js (lines 378–379):
if (CULT.MANTRA_SYSTEM && n.mantraState)
CULT.MANTRA_SYSTEM.stepMeditationResonance(W, n);
if (CULT.SUPPORT_ARTS && n.supportState && (W.t % 15 === 0))
CULT.SUPPORT_ARTS.stepFacilityAndDailySynergy(W, n);

// js/combat.js (lines 765, 811, 871, 916):
if (a && !a.foe && a.cult && a.mantraState) { ... }
if (d.defensiveArtState && d.cult) { ... }
if (!supNpc.supportState || !supNpc.cult) continue;
```

```
if (a && !a.foe && a.forbiddenArtState && a.cult) { ... }
```

Now trace where `n.mantraState`, `n.defensiveArtState`, `n.supportState`, and `n.forbiddenArtState` are created:

- `SIM.buildNpc` / `SIM.awaken`: **Does NOT initialize them (`undefined`)!**
- `AI.step` / `COG` / `COMBAT.startFight`: **Never calls `ensureState(n)` or `set*Posture(W, n, ...)`!**
- **The ONLY callers in the entire repository** are unit tests and **manual player-click buttons in `ui.js`!**
- **Dual Failure**:
    1. **Code Bug**: In an actual hands-off autonomous game, `n.mantraState`, `n.defensiveArtState`, `n.supportState`, and `n.forbiddenArtState` remain `undefined` forever on every single NPC in the world—meaning Mantra Auras, Defensive Kungfu Parries/Bodyblocks/Cicada Molt Escapes, Support Hall Synergies, and Forbidden Blood-Taboo Unsealing **never execute once**!
    2. **Genre Violation**: Relying on the player to click stance/posture/taboo buttons on 80 individual NPCs violates the core JianghuSect 2 rule: **the player cannot control or puppeteer NPCs!**

8.3 Autonomous Cognitive Stance & Seal Engine Blueprint (Replace Player Buttons With NPC Brain!)

2. **Auto-Initialize on Awakening & Hydration**:
    - Call `MANTRA_SYSTEM.ensureState(n)`, `DEFENSIVE_ARTS.ensureState(n)`, `SUPPORT_ARTS.ensureState(n)`, and `FORBIDDEN_ARTS.ensureState(n)` inside `SIM.awaken`, `M.retrain`, and `SIM.deserialize` for every awakened cultivator (`!!n.cult`).
2. **Convert UI Buttons to Read-Only Cognitive Telemetry**:
    - Remove player posture/seal-toggling commands from `ui.js` (keep stat/item sandbox editing only in `SECT.EDITOR`). Display the NPC's current autonomous stance and their `stanceWhyTh` reason string in the Character Sheet!
3. **Implement Deterministic Utility Evaluators (`CULT.evaluateAutonomousPostures(W, n, fightContext)`)**:
    - **Mantra Stance** (evaluated at start of `cultivate` / `consolidate` / `seclusion` and `COMBAT.startFight`):
        - If `(n.cult.toxic > 32 || n.h.state.heartDemon > 38)` and `caution > 40` -> **`SERENE_PURIFICATION`** (โคจรปราณชำระพิษและมารในใจ).

    - Else if `(n.cult.bn || ambition > 74)` and `hpFrac > 0.80` and `caution < 55` -> **`OVERDRIVE_COMBUSTION`** (เร่งเผาผลาญปราณทะลวงขีดจำกัด).
    - Else if `empathy > 62` or `n.rank >= 3` with nearby junior sectmates -> **`SHARED_AURA_CONDUIT`** (แผ่รัศมีปราณเกื้อหนุนศิษย์ร่วมสำนัก).
    - Else -> **`BALANCED_CIRCULATION`**.
- **Defensive Posture** (evaluated at `COMBAT.startFight` and when `hpFrac` crosses `0.45`):
    - If `hpFrac < 0.32` and `caution > courage` -> **`VOID_PHASE_EVASION`** (unlocks autonomous Cicada Molt survival escape!).
    - Else if `COMBAT.getUnitRow(n) === 'FRONT_ROW'` and `(loyalty > 62 || empathy > 62)` and allies present -> **`FORTRESS_BASTION_LOCK`** (ตั้งปราการคุ้มกันสหายแนวหลัง).
    - Else if `pride > 62 || aggression > 60` -> **`MIRROR_COUNTER_PARRY`** (ปัดป้องสวนคมดาบ).
    - Else -> **`FLOWING_GUARD`**.
- **Support Posture** (evaluated daily in `COG` and at `COMBAT.startFight`):
    - In peacetime when injured/sick sectmates exist or working in `infirmary`/`alchemy`/`herb_garden` -> **`MERIDIAN_HARMONY_PULSE`**.
    - During `raid` / `siege` / `guardian_combat` -> **`WAR_DRUM_CONDUCTION`**.
- **Forbidden Seal Mode** (evaluated dynamically per combat round when `n` actually knows a `FORBIDDEN` technique):
    - Default = **`SEALED_SAFE`**.
    - Respect Sect Policy (`W.sect.policy.strictness`) & `VOW_ASCETIC_PRECEPT` **unless** personality/desperation overrides it:
        - Compute `tabooDrive = (aggression - 50)*0.4 + (ambition - 50)*0.35 + (heartDemon)*0.5 - (integrity - 50)*0.6 - (caution - 50)*0.3 + (hpFrac < 0.25 ? 42 : 0) + (fightingBlackThreadNemesis ? 30 : 0) - (hasAsceticVow ? 45 : 0) - (sectForbidsTaboo ? (loyalty * 0.4) : 0)`.
        - If `tabooDrive >= 55` (e.g., a ruthless demonic cultivator, or a cornered disciple at `<20% HP` facing death against their master's killer!), the NPC **autonomously unseals** **`BLOOD_TABOO_UNLEASHED`** (and if they held `VOW_ASCETIC_PRECEPT`, they consciously shatter their vow in a dramatic chronicle-logged moment to survive!).

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9. [PATH-GDAO-09] Disembodied Monthly God-Scripts, -76% Pantheon Inversion, Split-Brain Runtime Stances & SHADOW_SAMURAI Zero-Qi Bugs

9.1 Exact Code Anchors

- `js/cultivation.js`: `CULT.GRAND_DAO_MECHANICS.monthlyAutonomousTick` (lines 2040–2138), `devourResourceForBody` (lines 2270–2282), `CULT.HIGHER_REALMS_PANTHEON` (lines 2104–2230), `CULT.DEMONIC_PANTHEON` (lines 2500–2515)
- `js/stats.js`: `compute` (lines 340–364) vs `getBaseFast` (lines 486–507)
- `js/cpeco.js`: `CPE.pickPurchase` (line 230)
- `js/visitors_clues.js`: `COMBAT.strike` Xiaoli Dagger hook (line 756) & `EFX_LEG_XIAOLI_TRUE_STRIKE` (line 631)

9.2 Root-Cause Code & Logic Defects

Defect 9A — `monthlyAutonomousTick` Is a Disembodied Top-Down God-Script That Violates NPC Identity & Location

In `cultivation.js:2046–2135`, once per month (`SIM.monthly`), `monthlyAutonomousTick` runs global hooks that:

1. Use `|| aliveCults[0]` fallbacks when no disciple in the sect practices `DEVOURING_FLESH`, `NETHER_GRAVE`, or Divine paths—**forcing `aliveCults[0]` (often an orthodox `XIAN` or `BUDDHA_DHARMA` Sect Master!) to devour random warehouse items (`grain`, `meal`, `ITEM_REPAIR_STONE_4`, Breakthrough Treasures) as "mutated flesh organs"** and channel necromantic grave echoes!
2. Execute world-altering rituals on NPCs **without checking `!n.away`, `!n.sealed`, `!(n.jail > W.t)`, or station proximity**—an imprisoned or expedition-away disciple telepathically anchors sect leylines and signs mercantile escrow contracts from 500 miles away!
3. Debit **`W.sect.stones`** (`40` stones for Cosmic Shrine, `160` stones for personal Demon Lord Blood Covenants in `signBloodCovenant`, `100` stones for Mercantile escrow, `50` stones for `FORTUNE_GAMBLER` wagers) **without Treasury/Master authorization**, allowing a Rank-0 outer disciple to gamble or embezzle public Sect Treasury stones for personal Demon pacts!

Defect 9B — CRITICAL `-68%` to `-82%` Stat Cripple on All 20 Higher-Realm Deities & 15 Demon Lords (`cultivation.js:2205–2228`)

`CULT.HIGHER_REALMS_PANTHEON.DEITIES` and `DEMON_LORDS` store bonus percentages as `0.18..0.32` (e.g., `PANGU.atkMult = 0.24`, `SHIVA_NATARAJA.matkMult = 0.28`, `VISHNU_PRESERVER.defMult = 0.24`), yet `formDeityPact` and `getUnitBlessingBonuses` compute `+(deity.atkMult - 1).toFixed(3)` -> **`-0.76` (`-76% ATK`)**!

- **Impact**: Every cultivator on `DIVINE_SUMMONER`, `DIVINE_POSSESSION`, `DIVINE_CHANNELER`, and `VOID_HERETIC_PACT` who forms their signature Deity Pact loses ~75% of their combat stats!

Defect 9C — Split-Brain `MECH_PUPPET` & `KARMA_SAMSARA` Runtime Stances + Missing Combat AI Switching (`stats.js`)

1. `ST.explain` (`compute`) and `ST.get` (`getBaseFast`) disagree on every single mode of `MECH_PUPPET` (`OVERDRIVE` falls into the defensive `else` branch in `getBaseFast`; bonuses still apply at `0%` integrity) and every single stance of `KARMA_SAMSARA` (`DEVA`, `ASURA`, `PRETA`, `NARAKA`, `HUMAN`, `ANIMAL`).
2. Furthermore, `COMBAT.fightStep` lacks autonomous tactical mode-switching for `MECH_PUPPET` (switching `AEGIS` <-> `SIEGE` <-> `SWARM` <-> `OVERDRIVE` based on construct `integrity`, `heat`, enemy count, and row) and `KARMA_SAMSARA` (rotating the Six-Paths Wheel stance based on `debtPool`, `hpFrac`, and target sin!).

Defect 9D — `SHADOW_SAMURAI` (`Zero-Qi Bushido`) Buys Useless Qi Pills & Fails Qi-Gated Blade Relics

Because `CULT.isZeroQiSamurai(n)` sets `qiMax = 0` and `n.h.state.qi = 0`:

1. **`CPE.pickPurchase` (`cpeco.js:230`)**: Evaluates `e.qiRestore && c && have(k) < 1 && n.rank >= 1` (`true`!), causing `SHADOW_SAMURAI` disciples to spend CP buying `P:BASIC_QI` Qi-recovery pills that restore `0 * 0.35 = 0` Qi! Similarly, `EXT.memberQuote` (`external_market.js:495`: `s.qi < CULT.qiMax(n) * 0.25` -> `0 < 0` is `false`, safe, **except** line 506 speculative medicine buying still buys `qiRestore` pills!).
2. **Blade Relic Failure (`visitors_clues.js:631, 756`)**: `LEG_WPN_XIAOLI_HEART_DAGGER` (`gearType: 'BLADE'`, legal Bushido weapon!) checks `needQi = Math.max(1, Math.round(qm * 0.04))` -> `1`, and `(attacker.h.state.qi || 0) >= 1` -> **`0 >= 1` (`false`)**, permanently disabling its proc for `SHADOW_SAMURAI`!
3. **Cultivation Tick (`ai.js:457`)**: Skips Qi regen for `SHADOW_SAMURAI` without regenerating `IAI_FOCUS` (`n.pathGauge.value`) or practicing `BLADE`/`SWORD`.

9.3 Architectural Remediation Blueprint for AI Developers

1. **Replace `monthlyAutonomousTick` God-Hand With Genuine Per-NPC Utility Actions (`AI.ACTS.grandDaoPractice`)**:
    - Remove all `|| aliveCults[0]` fallbacks immediately.
    - Convert path-specific world actions into a physical `AI.ACTS.pathRitual` (or per-domain `AI.ACTS`) where an eligible, present (`!n.away && !n.sealed && !(n.jail > W.t)`) cultivator on that specific path walks to the appropriate building (`shrine`, `ancestral`, `market`, `forge`, `herb_garden`, `ruin`), spends `60–120` ticks, pays personal `n.inv.stones` / personal materials (or requests formal officer approval via `VC.authorizeClueResourceUse`-style reserve & alignment checks for sect-wide buffs like `harmonizeSkyWeather` / `anchorDragonVeinNode`), and logs a causal `why` entry!
    - Restrict `devourResourceForBody` strictly to beast parts/organs (`MAT_BP_*`, `MAT_V222_BP_*`, surplus `W.sect.organs`), never grain, meals, repair stones, or breakthrough treasures.
2. **Fix `HIGHER_REALMS_PANTHEON` Multiplier Math & Hydrate Corrupted Saves**:
    - Use `const toPct = v => !v ? 0 : (v >= 1 ? +(v - 1).toFixed(3) : +v.toFixed(3));` in `formDeityPact` and `getUnitBlessingBonuses`, and repair saved `npc.deityPact` objects with negative `atkPct`/`matkPct`/`defPct`/`mdefPct`.
3. **Unify `MECH_PUPPET` & `KARMA_SAMSARA` Stat Helpers + Add Autonomous Mid-Combat Wheel/Mode Switching**:
    - Share one stat helper between `compute` and `getBaseFast`, gate construct bonuses on `integrity >= 25`, and add tactical stance/mode evaluation every `8` combat ticks in `COMBAT.fightStep` driven by NPC `courage`/`caution`/`rationality`, `hpFrac`, `integrity`, and `debtPool`.
4. **Make `SHADOW_SAMURAI` First-Class Across Economy & Relics**:
    - Block `qiRestore`-only pills for `CULT.isZeroQiSamurai(n)` in `CPE.pickPurchase`, `CPE.autoUse`, `CPE.fightUse`, `EXT.memberQuote`, and `COG.giveMedicine`.
    - Allow `CULT.isZeroQiSamurai(attacker)` to pay `IAI_FOCUS` (`8` gauge points) instead of `needQi` on Blade/Sword relic procs like `EFX_LEG_XIAOLI_TRUE_STRIKE`, and restore `IAI_FOCUS` + practice `BLADE`/`SWORD` during `A.cultivate.tick`.

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10. [PATH-FUT-10] Future-Proof Declarative PATH_BEHAVIOR_DESCRIPTOR Schema (Preventing Wave-5+ Drift Forever)

10.1 Why Hardcoded `if (c.path === ...)` Ladders Fail at Scale

Right now, adding or modifying a Major Cultivation Path requires editing **9 separate files** (`data_cult.js`/`data_martial.js`, `cultivation.js`, `ai.js`, `bottleneck.js`, `cpeco.js`, `stats.js`, `martial.js`, `combat.js`, `sect.js`). Whenever an AI model updates 6 of those 9 files, the new path silently breaks in the other 3 (e.g., cultivating at the blacksmith forge, breaking through at the god shrine, gaining scholar wisdom instead of body strength, or receiving `+matk` on a zero-Qi physical swordsman).

10.2 Target Single-Source-of-Truth Schema (`CULT.PATH_DESCRIPTORS[pathId]`)

Refactor path-specific hooks to read from a single validated registry `CULT.PATH_DESCRIPTORS` (requiring 100% coverage in `tools/validate_data_schema.js` for every key in `CULT.PATHS`):

JavaScript

```
// Canonical schema required for every Major Path (all 43 current paths + any
future path):
{
  id: 'SHADOW_SAMURAI',
  resourceModel: 'ZERO_QI_FOCUS',          // 'QI' | 'ZERO_QI_FOCUS' | 'FAITH' |
'BLOOD_QI'
  archetypeRole: 'PHYSICAL_BURST_BLADE',   // governs default AI & combat fallback
  stations: {
    cultivate: [
      { bkey: 'PEAK_SELF', kind: 'practice', mult: 1.38 },
      { bkey: 'training_ground', kind: 'dummy', mult: 1.28 },
      { bkey: 'meditation_hall', kind: 'meditate', mult: 1.15 }
    ],
    breakthrough: { bkey: 'training_ground', kind: 'dummy' },
    consolidate:  { bkey: 'training_ground', kind: 'dummy' },
    paidTraining: { bkey: 'training_ground', kind: 'dummy' }
  },
  cultivateTick: {
    practiceSkills: [{ skill: 'EQUIPPED_WEAPON', weight: 0.7 }, { skill:
'MOVEMENT', weight: 0.3 }],
    trainBody: ['reflex', 'agility', 'explosiveness'],
    trainMind: ['composure'],
```

```
    heartDemonDeltaPer30: -0.025,          // Bushido mindfulness calms heart
demons
    gaugeRegenPerTick: 0.8
  },
  scaling: {
    hpGrowthPerNormStage: 0.06,            // replaces hardcoded ternary in
CULT.hpMult
    mantraPerTier: { p: { atk: 0.05, def: 0.03 }, f: { crit: 1.2, move: 1.0 } },
// replaces stats.js hardcoded ladder
    combatOffenseBaseMult: 1.10,
    combatDefenseTakenMult: 0.94
  },
  specialElementUnlock: null,              // 'ASTRAL' | 'BLOOD_DEMON' | null
  worldRitualAct: null                     // optional physical AI.ACTS spec
(replaces monthlyAutonomousTick)
}
```

10.3 Mandatory End-to-End Verification Suite Spec (For Part 2 Fixes)

When GPT or Gemini implements Part 2 remediations, it must add `test/v270_paths_subpaths_part2_parity_and_autonomy_suite.js` asserting:

1. **`test_pantheon_deity_and_demon_pacts_positive_stats`**: Form pacts across all 20 Deities & 15 Demon Lords; assert `STATS.get(n)` increases `atk`/`matk`/`def`/`mdef` (zero `-0.76` inversions!) and `STATS.get(n)` matches `STATS.explain(n).v` within `<= 1` rounding unit across all `MECH_PUPPET` modes and `KARMA_SAMSARA` stances.
2. **`test_stage1_track_cannot_equip_tier7_divine_relic`**: Set a Stage-1 disciple's `c.track.ordinal = 8`; assert `CULT.effectiveTechStage(n) <= 2` and `MARTIAL.gearUsable(n, tier7Relic) === false`.
3. **`test_all_43_paths_subpath_traits_3_6_9_reachable`**: At Apex stage (`maxStage` `4`, `5`, `8`, or `9`), verify `curOrd` unlocks all 3 `subpathTraits` (`reqRealm` 3, 6, 9) even if `c.track.ordinal === 1`.
4. **`test_awakened_peak_stations_used_by_cultivate_and_practice`**: Assign a disciple to `PEAK_AZURE_SWORD`, saturate central `meditation_hall`, run `AI.decide(W, n)` for `cultivate`, and verify the disciple routes to and occupies a physical `'meditate'` station on `PEAK_AZURE_SWORD`.
5. **`test_era5_postures_and_forbidden_unseal_100pct_autonomous`**: Spawn freshly awakened NPCs (`SIM.create`), execute **zero** UI/player posture calls, run live `SIM.tick` + `COMBAT.startFight`, and verify: (a) poisoned/high-heartDemon NPC autonomously adopts `SERENE_PURIFICATION`, (b) loyal vanguard adopts `FORTRESS_BASTION_LOCK` and triggers defensive mitigation, and (c) desperate `<20% HP` low-integrity Demon cultivator autonomously unseals `BLOOD_TABOO_UNLEASHED` in combat.

6. **`test_zero_disembodied_god_hand_cannibalism`**: Run `SIM.monthly` for 12 months in a pure `XIAN` sect stocked with `grain`, `ITEM_REPAIR_STONE_3`, and Breakthrough Treasures; assert `devouredOrgansCount === 0`, zero repair stones/grain eaten by `DEVOURING_FLESH`, and zero unauthorized `W.sect.stones` spent on Demonic Covenants by outer disciples.
7. **`test_shadow_samurai_zero_qi_economy_hygiene`**: Give a `SHADOW_SAMURAI` 500 CP in a sect stocked with `P:BASIC_QI`; assert `CPE.pickPurchase` never selects `qiRestore`-only pills, and wielding `LEG_WPN_XIAOLI_HEART_DAGGER` triggers its proc by spending `IAI_FOCUS`.